

Shashank Shekhar Shukla

shuklashashankshekhar863@gmail.com | github.com/Wodlfvllf | [Portfolio](#)

Education

- **Indian Institute of Technology (IIT) Dhanbad** Dhanbad, India
B.Tech. in MME, Minor in CSE, GPA: 8.14/10.0 Dec 2021 – May 2025
 - **Relevant Coursework:** Deep Learning, Machine Learning, Parallel Computing, Data Structures & Algorithms

Technical Skills

Programming Languages: Python, C++, Bash, SQL
ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers
Distributed Training: Data Parallelism, Tensor Parallelism, Pipeline Parallelism, PyTorch DDP/FSDP, NCCL
Systems & Infra: CUDA basics, Docker, Git, Linux, AWS, GCP, Weights & Biases
LLMs & RL: LLaMA, Qwen, LoRA, PEFT, QLoRA, GRPO, RLHF-style post-training, RAG

Open Source Research & Systems Work

- **grpo-composer – Modular RL Post-Training Framework:** Built a modular RL post-training framework on top of veRL for experimenting with GRPO-style algorithms. Implemented pluggable reward functions, advantage estimators, clipping strategies, regularizers, aggregation schemes, and reward transforms. Integrated distributed rollout and training components using PyTorch-based infrastructure. Designed the system to support rapid experimentation without modifying the trainer core. [GitHub](#)
- **QuintNet – PyTorch Framework for Distributed Training:** Built a PyTorch framework for data, tensor, pipeline, sequence, and expert parallelism. Implemented distributed training support for GPT-style LLMs, vision-language models, and Mixture-of-Experts architectures. Designed communication-aware training components for multi-GPU scaling and model partitioning. [GitHub](#)
- **AMD AI Premier League – 1st Place Winner:** Built a dual-agent RL system where one agent generates logic-based MCQs and another solves them using reasoning. Fine-tuned Qwen3-4B and used GRPO with reward shaping, KL regularization, and self-play. Won 1st place among 500+ teams at IISc Bangalore. [GitHub](#)
- **High-Performance CUDA Kernels Library:** Built CUDA kernels for LLM inference, including FlashAttention variants, fused operations, and memory-efficient matrix kernels. Focused on throughput and memory optimization on modern GPUs. [GitHub](#)

Experience

- **Microsoft** Hyderabad, India
Applied Scientist Feb 2026 – Present
 - Developing transformer-based document classification systems for Microsoft Purview, including detection of sensitive content such as IP, source code, and confidential data.
 - Built synthetic and real-world dataset pipelines to improve coverage on ambiguous and edge-case examples.
 - Working on low-latency model improvements through structured pruning, linear probing, and architecture changes under production constraints.
 - Built evaluation and verification pipelines for scenario testing and counterfactual checks.
 - Collaborating with cross-functional teams to integrate ML models into enterprise systems with accuracy, safety, and compliance requirements.
- **FAMPay (YC S19) – FinTech** Bengaluru, India
Data Scientist Aug 2025 – Present
 - Built ETL pipelines in ClickHouse SQL for user verification data used in fraud detection and risk checks.
 - Built a name matching system using phonetic similarity, edit distance, and string matching, reducing false positives by 15% in production.
 - Built a graph-based address matching system using relational and structural similarity, reducing false positives by 25%.
 - Developed a churn prediction pipeline using TF-IDF, clustering, and entity extraction to identify user behavior patterns.
- **Google Summer of Code (ML4SCI at CERN)** Remote
Machine Learning Engineer May 2024 – Sep 2024
 - Built an end-to-end ML pipeline for 3.8M+ detector images from the CMS experiment.
 - Developed a self-supervised MAE + ViT model that achieved 98.3% AUC with 99.9% parameter reduction.
 - Implemented FSDP-based distributed training across A100 GPUs for scalable pretraining.
 - Released a reproducible pipeline for the scientific ML community.

- **Swiggy**

Data Scientist Intern

Remote

Jan 2025 – Jul 2025

- Built a pipeline to process 600K+ restaurant images using GPT-4o and Gemini APIs for structured data extraction.
- Designed a distributed inference pipeline with PySpark and Delta Lake to scale API usage and control cost.
- Fine-tuned embedding models with triplet loss on multi-source data to improve retrieval quality.
- Deployed a semantic search system that increased conversion for dineout recommendations by 15%.
- Implemented a hybrid recommendation system to handle cold-start cases using content and collaborative signals.

- **Tata Consultancy Services (TCS) Research**

Machine Learning Engineer Intern

Pune, India

May 2024 – Jul 2024

- Built an LLM-based query interface for natural-language access to industrial datasets.
- Fine-tuned Phi-3, LLaMA2, and OpenELM, improving F1 by 292% through domain adaptation.
- Built a transformer-based anomaly detection pipeline for manufacturing quality control.

Competitions & Recognition

- **Google Summer of Code 2024** – Selected for a competitive ML research internship.
- **Kaggle Competition Expert** – 1 Silver Medal, 3 Bronze Medals across multiple ML domains.
 - Silver: LLM Science Exam (75th/2800) – retrieval-augmented ensemble with DeBERTa variants
 - Bronze: NeurIPS LuxAI Season 3 (60th/1500) – multi-agent RL with 16 cooperating agents using IMPALA + V-trace
 - Bronze: UBC-OCEAN Cancer Classification (83rd/1600) – whole slide image classification using Multiple Instance Learning
- **AMD AI Hackathon Winner** – 1st place among 500+ teams at IISc Bangalore.
- **Amazon ML Challenge** – 28th rank in a large-scale ML competition.
- **GATE 2025 Data Science** – All India Rank 1503.